

Problem-Solution Fit

Date	06 July 2026
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas :

1. CUSTOMER SEGMENTATION (CS) Residents in flood-prone areas, Farmers, Disaster Management Authorities, Local Government Officials	6. CUSTOMER LIMITATIONS Lack of timely information, limited access to advanced prediction systems, low awareness, internet connectivity issue...	5. AVAILABLE SOLUTIONS Pros: Weather forecasts, government alerts, news updates. Cons: May be delayed, less localized, limited prediction accuracy.
2. PROBLEMS / PAINS Sudden floods, lack of early warnings, property damage, crop loss, loss of lives, inaccurate prediction methods	9. PROBLEM ROOT Climate change, unpredictable rainfall, poor drainage systems, deforestation, rapid urbanization, lack of accurate flood prediction systems.	7. BEHAVIOUR Regularly checks weather forecasts, follows news and government advisories, prepares during heavy rainfall, evacuates when alerts are issued.
3. TRIGGERS TO ACT Heavy rainfall, rising water levels, weather alerts, monsoon season, flood warnings	10. YOUR SOLUTION Develop an AI-powered Flood Prediction System using Machine Learning (XGBoost) and Flask that analyzes weather parameters and provides quick, reliable flood risk predictions through an easy-to-use web application.	8. CHANNELS OF BEHAVIOUR ONLINE Website, Mobile apps, SMS, Social media.
4. EMOTIONS Before: Fear, anxiety, uncertainty. After: Relief, confidence, preparedness, improved safety.		OFFLINE TV, Radio, Local announcements, Disaster management teams.

